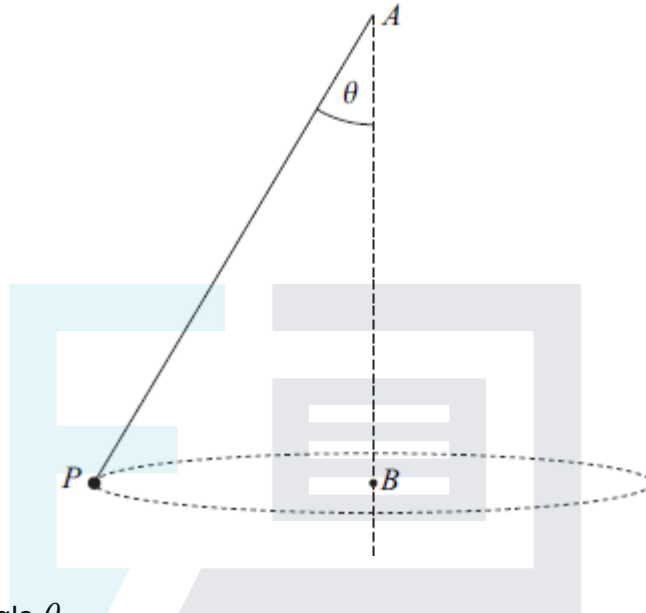


Q1.

A light inextensible string has one end attached to a particle, P , of mass 2 kg. The other end of the string is attached to the fixed point A . The point A is vertically above the point B . The particle moves at a constant speed in a horizontal circle of radius 0.8 m and centre B . The tension in the string is 34 N.

The string is inclined at an angle θ to the vertical, as shown in the diagram.



- (a) Find the angle θ .

(3)

- (b) Find the speed of the particle.

(3)

- (c) Find the time taken for the particle to make one complete revolution.

(2)

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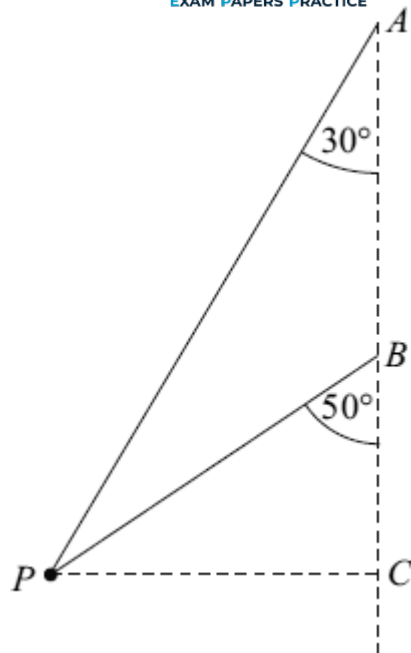
(Total 8 marks)

Q2.

Two light inextensible strings each have one end attached to a particle, P , of mass 4 kg. The other ends of the strings are attached to the fixed points A and B . The point A is vertically above the point B .

The particle moves at a constant speed in a horizontal circle. The centre, C , of this circle is directly below the point B . The two strings are inclined at 30° and 50° to the vertical, as shown in the diagram. Both strings are taut.

As the particle moves in the horizontal circle, the tension in the string BP is 20 N.

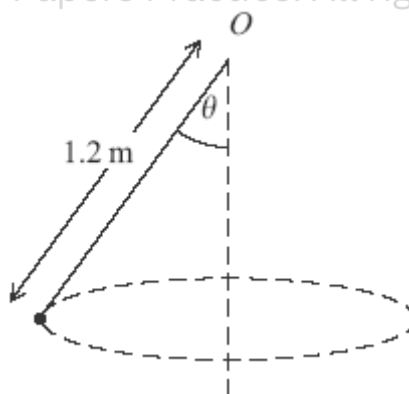


- (a) Find the tension in the string AP . (4)
- (b) The speed of the particle is 5 m s^{-1} .
Find the length of CP , the radius of the horizontal circle. (4)
- (Total 8 marks)

Q3.

A particle, of mass 4 kg , is attached to one end of a light inextensible string of length 1.2 metres . The other end of the string is attached to a fixed point O . The particle moves in a horizontal circle at a constant speed. The angle between the string and the vertical is θ .

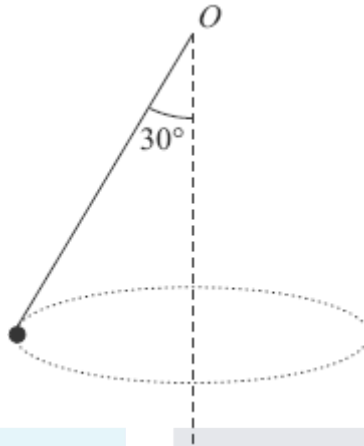
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- (a) Find the radius of the horizontal circle in terms of θ . (1)
- (b) The angular speed of the particle is $5 \text{ radians per second}$. Find θ . (6)
- (Total 7 marks)

Q4.

A particle, of mass 6 kg, is attached to one end of a light inextensible string. The other end of the string is attached to the fixed point O . The particle is set in motion, so that it moves in a horizontal circle at constant speed, with the string at an angle of 30° to the vertical. The centre of this circle is vertically below O .



The particle moves in a horizontal circle with an angular speed of 40 revolutions per minute.

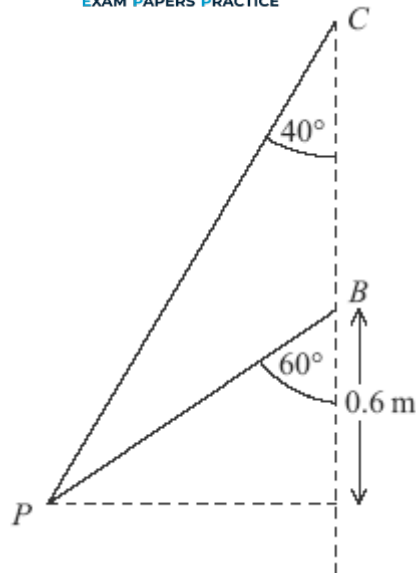
- (a) Show that the angular speed of the particle is $\frac{4\pi}{3}$ radians per second. (2)
- (b) Show that the tension in the string is 67.9 N, correct to three significant figures. (3)
- (c) Find the radius of the horizontal circle. (4)

(Total 9 marks)

Q5.

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Two light inextensible strings each have one end attached to a particle, P , of mass 6 kg. The other ends of the strings are attached to the fixed points B and C . The point C is vertically above the point B . The particle moves, at constant speed, in a horizontal circle, with centre 0.6 m below point B , with the strings inclined at 40° and 60° to the vertical, as shown in the diagram. Both strings are taut.



- (a) As the particle moves in the horizontal circle, the tensions in the two strings are equal.

Show that the tension in the strings is 46.4 N, correct to three significant figures.

(4)

- (b) Find the speed of the particle.

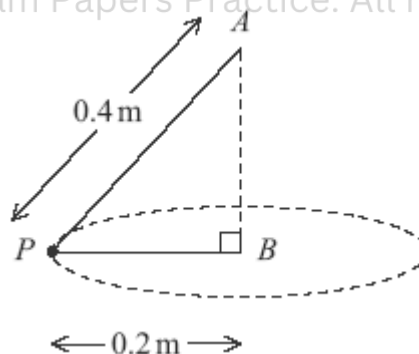
(4)

(Total 8 marks)

Q6.

Two light inextensible strings, of lengths 0.4 m and 0.2 m, each have one end attached to a particle, P , of mass 4 kg. The other ends of the strings are attached to the points A and B respectively. The point A is vertically above the point B . The particle moves in a horizontal circle, centre B and radius 0.2 m, at a speed of 2 m s^{-1} . The particle and strings are shown in the diagram.

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- (a) Calculate the magnitude of the acceleration of the particle.

(2)

- (b) Show that the tension in string PA is 45.3 N, correct to three significant figures.

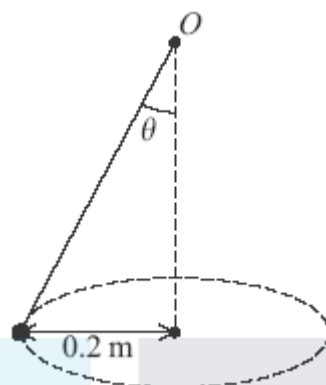
(4)

- (c) Find the tension in string PB .

(3)

Q7.

A particle is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point O . The particle is set into motion, so that it describes a horizontal circle whose centre is vertically below O . The angle between the string and the vertical is θ , as shown in the diagram.



- (a) The particle completes 40 revolutions every minute.

Show that the angular speed of the particle is $\frac{4\pi}{3}$ radians per second.

(2)

- (b) The radius of the circle is 0.2 metres.

Find, in terms of π , the magnitude of the acceleration of the particle.

(2)

- (c) The mass of the particle is m kg and the tension in the string is T newtons.

(i) Draw a diagram showing the forces acting on the particle.

(1)

(ii) Explain why $T \cos \theta = mg$.

(1)

(iii) Find the value of θ , giving your answer to the nearest degree.

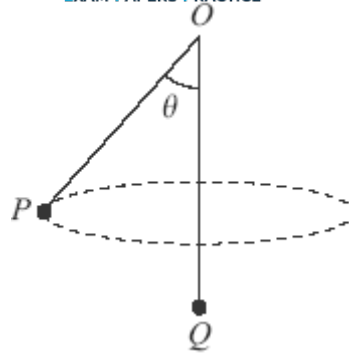
(5)

(Total 11 marks)

Q8.

A particle, P , of mass 3 kg is attached to one end of a light inextensible string. The string passes through a smooth fixed ring, O , and a second particle, Q , of mass 5 kg is attached to the other end of the string. The particle Q hangs at rest vertically below the ring and the particle P moves with speed 4 m s^{-1} in a horizontal circle, as shown in the diagram.

The angle between OP and the vertical is θ .



(a) Explain why the tension in the string is 49 N.

(2)

(b) Find θ .

(3)

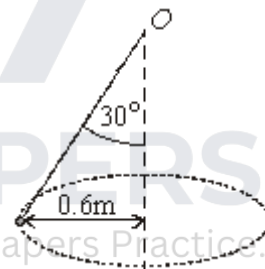
(c) Find the radius of the horizontal circle.

(4)

(Total 9 marks)

Q9.

A particle, of mass 2 kg, is attached to one end of a light inextensible string. The other end is fixed to the point O . The particle is set into motion, so that it describes a horizontal circle of radius 0.6 metres, with the string at an angle of 30° to the vertical. The centre of the circle is vertically below O .



(a) Show that the tension in the string is 22.6 N, correct to three significant figures.

(3)

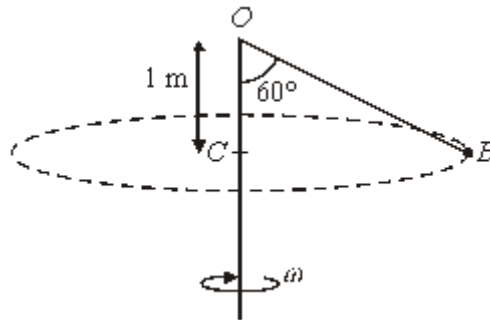
(b) Find the speed of the particle.

(4)

(Total 7 marks)

Q10.

The diagram shows a light inextensible cord that has one end O attached to the top of a fixed vertical pole.



A ball B of mass 0.4 kg is attached to the other end of the cord. The ball rotates around the pole with constant angular speed ω radians per second. It moves in a horizontal circle with centre C at a distance 1 metre vertically below O . The angle between OB and OC is 60° .

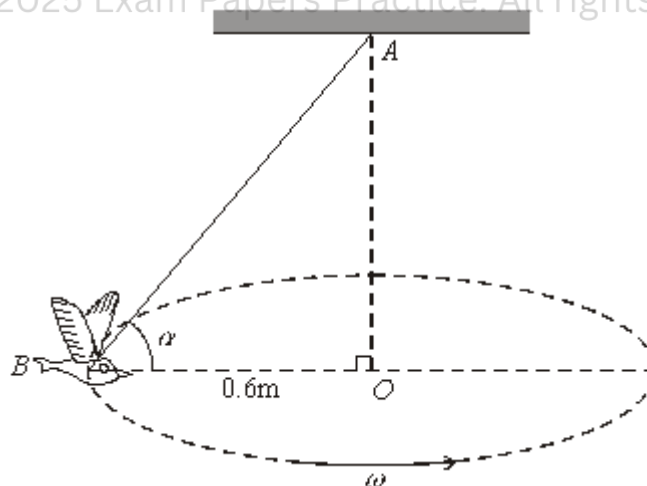
- Show that the tension in the cord is 7.84 newtons . (3)
 - Show that $\omega = 3.13$, correct to three significant figures. (5)
 - Show that the ball takes approximately two seconds to travel once around the pole. (1)
- (Total 9 marks)

Q11.

A toy bird, of mass 0.25 kg , is fixed to one end, B , of a light inextensible string. The other end, A , of the string is attached to a fixed point on a ceiling. The bird is set in motion, so that it describes a horizontal circle of radius 0.6 m . The centre of this circle is O , which is vertically below A , as shown in the diagram. The angle ABO is α .

The bird moves with a constant angular speed of ω radians per second.

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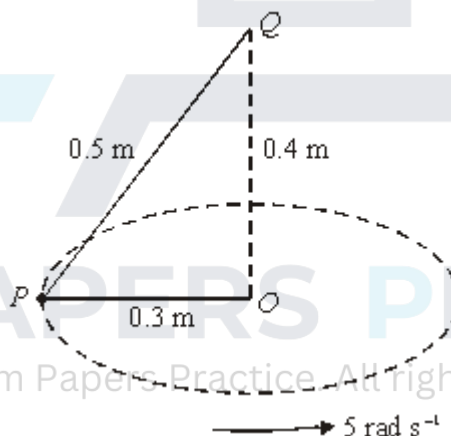
The bird completes one full revolution every 1.5 seconds .

- Find ω , leaving your answer in terms of π . (2)

- (b) (i) Determine the magnitude of the acceleration of the bird. (2)
- (ii) Show the direction of this acceleration on a diagram. (1)
- (c) (i) Draw a diagram to show the forces acting on the bird. (1)
- (ii) Find the angle α , giving your answer to the nearest degree. (6)
- (d) State **one** modelling assumption used in this question. (1)
- (Total 13 marks)**

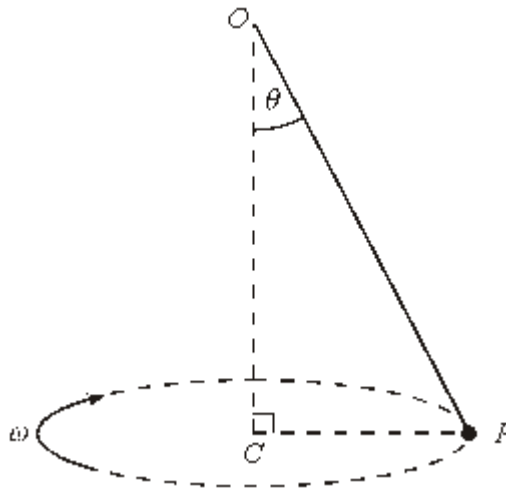
Q12.

A particle of mass 0.4 kg is attached at the point P to two light strings, QP and OP . The points O and Q are fixed with Q at a distance of 0.4 m vertically above O . The string QP is inextensible and of length 0.5 m . The string OP is elastic and of natural length 0.2 m and stiffness $k \text{ N m}^{-1}$. The particle moves in a horizontal circle, centre O and radius 0.3 m , at a constant angular speed of 5 rad s^{-1} .



- (a) Draw a diagram showing the forces acting on the particle. (1)
- (b) Show that the tension in the string QP is 4.9 N . (3)
- (c) Write down, in terms of k , the tension in the string OP . (1)
- (d) Show that $k = 0.6$. (5)
- (e) Find the elastic potential energy stored in the string OP . (2)
- (Total 12 marks)**

Q13.



A particle, P , of mass m is attached by a light, inextensible thread of length l to a fixed point O . The particle moves with constant angular speed ω in a horizontal circle with centre C . The point C is vertically below O . The thread is inclined to the vertical at an angle θ , as shown in the diagram.

- (a) Find an expression for the tension in the thread. Give your answer in terms of m , l and ω .

(4)

- (b) Show that $\cos \theta = \frac{g}{l\omega^2}$.

(3)

- (c) The greatest tension that the thread can withstand without breaking is 16 N.

In the case when $l = 0.4$ metres and $m = 0.1$ kg:

- (i) show that $\omega \leq 20$ radians per second;

(2)

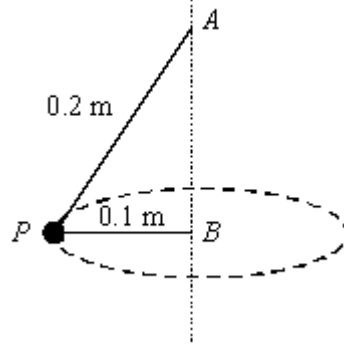
- (ii) find the greatest possible value of θ , giving your answer to the nearest degree.

(2)

(Total 11 marks)

Q14.

A particle, P , of mass 2 kg, is attached to two strings of lengths 0.2 metres and 0.1 metres. The strings are fixed to the particle and to the points A and B respectively. The point A is directly above the point B . The particle describes a horizontal circle, centre B and radius 0.1 metres, at a speed of 4 m s^{-1} . The particle and strings are shown in the diagram.



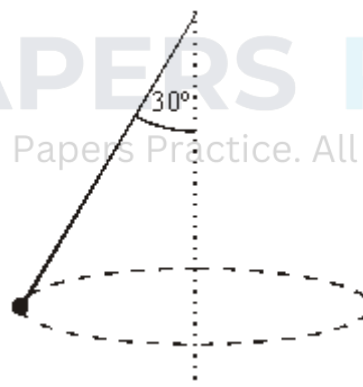
- (a) Calculate the magnitude of the acceleration of the particle. (2)
 - (b) Find the tension in the upper string. (4)
 - (c) Find the tension in the lower string (5)
- (Total 11 marks)**

Q15.

A conical pendulum consists of a small sphere of mass 3 kg attached to the end of a light, inextensible string of length 0.6 metres. The sphere moves in a horizontal circle at a constant speed.

Model the sphere as a particle.

- (a) The string is inclined at an angle of 30° to the vertical, as shown in the diagram.



- (i) Find the tension in the string. (3)
 - (ii) Show that the angular speed of the sphere is 4.34 radians per second, to three significant figures. (4)
- (b) The angular speed of the particle is now doubled. Find the new angle between the string and the vertical, giving your answer to the nearest degree. (6)
- (Total 13 marks)**